

# Customer Shopping Behavior Analysis

## 1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

## 2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

## 3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

|        | Customer ID | Age         | Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Shipping Type | Discount Applied |
|--------|-------------|-------------|--------|----------------|----------|-----------------------|----------|------|-------|--------|---------------|---------------------|---------------|------------------|
| count  | 3900.000000 | 3900.000000 | 3900   | 3900           | 3900     | 3900.000000           | 3900     | 3900 | 3900  | 3900   | 3863.000000   | 3900                | 3900          | 39               |
| unique | NaN         | NaN         | 2      | 25             | 4        | NaN                   | 50       | 4    | 25    | 4      | NaN           | 2                   | 6             |                  |
| top    | NaN         | NaN         | Male   | Blouse         | Clothing | NaN                   | Montana  | M    | Olive | Spring | NaN           | No                  | Free Shipping |                  |
| freq   | NaN         | NaN         | 2852   | 171            | 1737     | NaN                   | 96       | 1755 | 177   | 999    | NaN           | 2847                | 675           | 22               |
| mean   | 1950.500000 | 44.088462   | NaN    | NaN            | NaN      | 59.764359             | NaN      | NaN  | NaN   | NaN    | 3.750065      | NaN                 | NaN           | N                |
| std    | 1125.677353 | 15.207589   | NaN    | NaN            | NaN      | 23.685362             | NaN      | NaN  | NaN   | NaN    | 0.716963      | NaN                 | NaN           | N                |
| min    | 1.000000    | 18.000000   | NaN    | NaN            | NaN      | 20.000000             | NaN      | NaN  | NaN   | NaN    | 2.500000      | NaN                 | NaN           | N                |
| 25%    | 975.750000  | 31.000000   | NaN    | NaN            | NaN      | 39.000000             | NaN      | NaN  | NaN   | NaN    | 3.100000      | NaN                 | NaN           | N                |
| 50%    | 1950.500000 | 44.000000   | NaN    | NaN            | NaN      | 60.000000             | NaN      | NaN  | NaN   | NaN    | 3.000000      | NaN                 | NaN           | N                |
| 75%    | 2925.250000 | 57.000000   | NaN    | NaN            | NaN      | 81.000000             | NaN      | NaN  | NaN   | NaN    | 4.400000      | NaN                 | NaN           | N                |
| max    | 3900.000000 | 70.000000   | NaN    | NaN            | NaN      | 100.000000            | NaN      | NaN  | NaN   | NaN    | 5.000000      | NaN                 | NaN           | N                |

| Discount Applied | Promo Code Used | Previous Purchases | Payment Method | Frequency of Purchases |
|------------------|-----------------|--------------------|----------------|------------------------|
| 3000             | 3000            | 3000.000000        | 3000           | 3000                   |
| 2                | 2               | NaN                | 6              | 7                      |
| No               | No              | NaN                | PayPal         | Every 3 Months         |
| 2223             | 2223            | NaN                | 677            | 584                    |
| NaN              | NaN             | 25.351538          | NaN            | NaN                    |
| NaN              | NaN             | 14.447125          | NaN            | NaN                    |
| NaN              | NaN             | 1.000000           | NaN            | NaN                    |
| NaN              | NaN             | 13.000000          | NaN            | NaN                    |
| NaN              | NaN             | 25.000000          | NaN            | NaN                    |
| NaN              | NaN             | 38.000000          | NaN            | NaN                    |
| NaN              | NaN             | 50.000000          | NaN            | NaN                    |

- **Missing Data Handling:** Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
  - Created **age\_group** column by binning customer ages.
  - Created **purchase\_frequency\_days** column from purchase data.
- **Data Consistency Check:** Verified if **discount\_applied** and **promo\_code\_used** were redundant; dropped **promo\_code\_used**.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

## 4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in SQL Server to answer key business questions about customer purchasing patterns, discount effectiveness, product performance, and segmentation. Below are the findings from each analysis.

### 1. Revenue by Gender (Q1)

**Analysis:** Total revenue and percentage contribution by gender.

|   | gender | total_revenue | total_customers | revenue_percentage |
|---|--------|---------------|-----------------|--------------------|
| 1 | Male   | 157890        | 2652            | 67.74              |
| 2 | Female | 75191         | 1248            | 32.26              |

## 2. High-Spending Discount Users (Q2)

Analysis: Customers who used a discount but still spent more than their category's average purchase amount.

|    | rank_in_category | category    | customer_id | age | age_group   | purchase_amount | avg_by_category | amount_above_average |
|----|------------------|-------------|-------------|-----|-------------|-----------------|-----------------|----------------------|
| 1  | 1                | Accessories | 194         | 36  | Adult       | 100             | 59              | 41                   |
| 2  | 2                | Accessories | 244         | 25  | Young Adult | 100             | 59              | 41                   |
| 3  | 3                | Accessories | 249         | 47  | Midle-aged  | 100             | 59              | 41                   |
| 4  | 1                | Clothing    | 519         | 24  | Young Adult | 100             | 60              | 40                   |
| 5  | 2                | Clothing    | 582         | 32  | Adult       | 100             | 60              | 40                   |
| 6  | 3                | Clothing    | 456         | 54  | Midle-aged  | 100             | 60              | 40                   |
| 7  | 1                | Footwear    | 1209        | 20  | Young Adult | 100             | 60              | 40                   |
| 8  | 2                | Footwear    | 616         | 67  | Senior      | 100             | 60              | 40                   |
| 9  | 3                | Footwear    | 205         | 24  | Young Adult | 100             | 60              | 40                   |
| 10 | 1                | Outerwear   | 1480        | 48  | Midle-aged  | 100             | 57              | 43                   |
| 11 | 2                | Outerwear   | 1457        | 59  | Senior      | 100             | 57              | 43                   |
| 12 | 3                | Outerwear   | 43          | 20  | Young Adult | 100             | 57              | 43                   |

## 3. Top 5 Products by Average Review Rating (Q3)

Analysis: Highest-rated products based on customer reviews.

|   | item_purchased | category    | avg_product_rating |
|---|----------------|-------------|--------------------|
| 1 | Gloves         | Accessories | 3,86               |
| 2 | Sandals        | Footwear    | 3,84               |
| 3 | Boots          | Footwear    | 3,82               |
| 4 | Hat            | Accessories | 3,8                |
| 5 | Handbag        | Accessories | 3,78               |

## 4. Average Purchase Amount: Express vs. Standard Shipping (Q4)

Analysis: Comparison of average purchase amounts across shipping types, per category.

|   | category    | shipping_type | avg_purchase_amount |
|---|-------------|---------------|---------------------|
| 1 | Accessories | Express       | 61                  |
| 2 | Clothing    | Express       | 61                  |
| 3 | Footwear    | Express       | 59                  |
| 4 | Outerwear   | Express       | 56                  |
| 5 | Accessories | Standard      | 58                  |
| 6 | Clothing    | Standard      | 60                  |
| 7 | Footwear    | Standard      | 57                  |
| 8 | Outerwear   | Standard      | 52                  |

## 5. Subscribers vs. Non-Subscribers Spending (Q5)

Analysis: Total revenue and average spend comparison.

|   | customer_type  | total_customers | total_revenue | avg_spend |
|---|----------------|-----------------|---------------|-----------|
| 1 | subscriber     | 1053            | 62645         | 59        |
| 2 | non_subscriber | 2847            | 170436        | 59        |

## 6. Top 5 Products with Highest Discount Usage Percentage (Q6)

Analysis: Products most frequently purchased with a discount, along with revenue contribution from discounted sales.

|   | item_purchased | total_orders | discount_orders | discount_order_pct | total_revenue | pct_of_product_revenue_from_discount | discount_revenue_share_of_total |
|---|----------------|--------------|-----------------|--------------------|---------------|--------------------------------------|---------------------------------|
| 1 | Pants          | 171          | 81              | 47.37              | 10090         | 47.94                                | 2.08                            |
| 2 | Sweater        | 164          | 79              | 48.17              | 9462          | 48.72                                | 1.98                            |
| 3 | Hat            | 154          | 77              | 50.00              | 9375          | 49.02                                | 1.97                            |
| 4 | Coat           | 161          | 79              | 49.07              | 9275          | 48.19                                | 1.92                            |
| 5 | Dress          | 166          | 75              | 45.18              | 10320         | 42.33                                | 1.87                            |

## 7. Customer Segmentation: New, Returning, Loyal (Q7)

Analysis: Segmenting customers based on number of previous purchases.

|   | customer_segment | Number of Customers |
|---|------------------|---------------------|
| 1 | New              | 83                  |
| 2 | Returning        | 701                 |
| 3 | Loyal            | 3116                |

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### 8. Top 3 Most Purchased Products per Category (Q8)

Analysis: Best-selling items within each product category.

|    | item_rank | category    | item_purchased | total_orders |
|----|-----------|-------------|----------------|--------------|
| 1  | 1         | Accessories | Jewelry        | 171          |
| 2  | 2         | Accessories | Belt           | 161          |
| 3  | 3         | Accessories | Sunglasses     | 161          |
| 4  | 1         | Clothing    | Blouse         | 171          |
| 5  | 2         | Clothing    | Pants          | 171          |
| 6  | 3         | Clothing    | Shirt          | 169          |
| 7  | 1         | Footwear    | Sandals        | 160          |
| 8  | 2         | Footwear    | Shoes          | 150          |
| 9  | 3         | Footwear    | Sneakers       | 145          |
| 10 | 1         | Outerwear   | Jacket         | 163          |
| 11 | 2         | Outerwear   | Coat           | 161          |

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### 9. Repeat Buyers & Subscription Likelihood (Q9)

Analysis: Among customers with >5 previous purchases, how many are subscribers?

|   | subscription_status | repeat_buyers |
|---|---------------------|---------------|
| 1 | Yes                 | 958           |
| 2 | No                  | 2518          |

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### 10. Revenue Contribution by Age Group (Q10)

Analysis: Total revenue generated per customer age group.

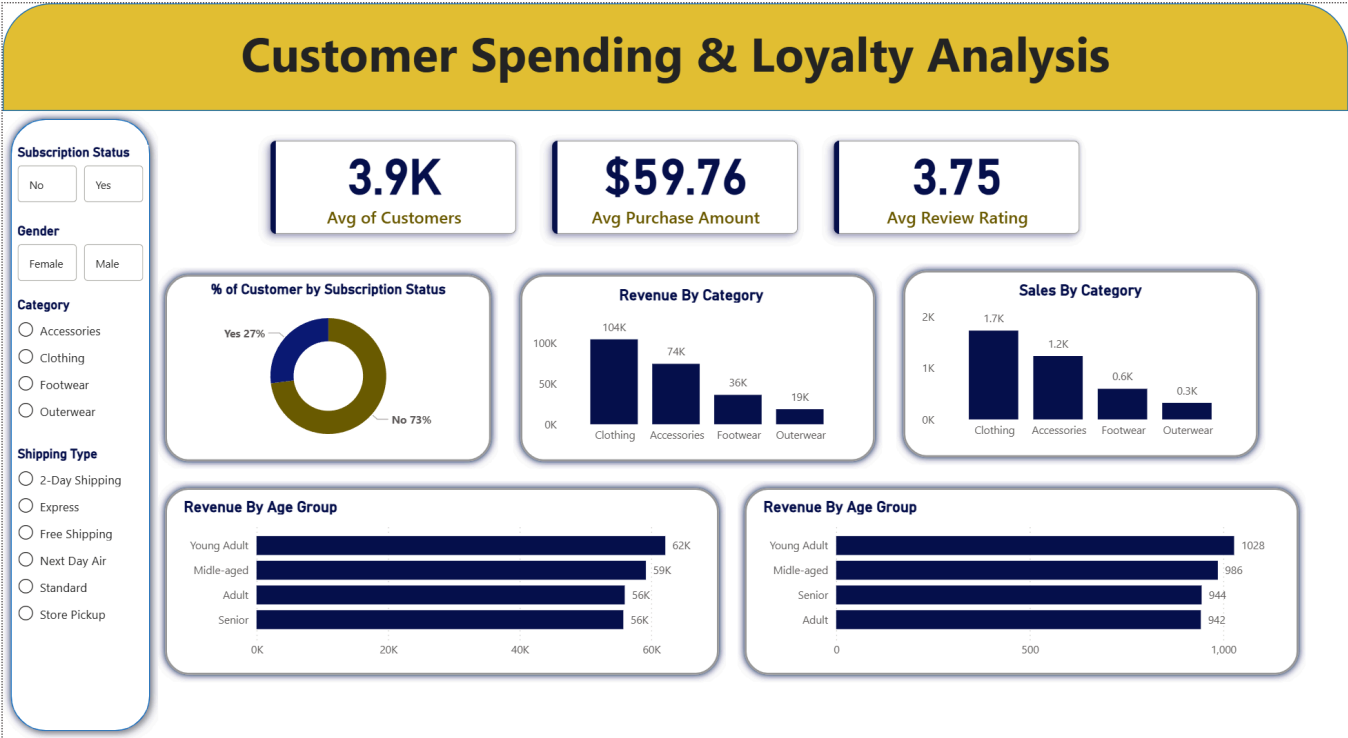
|   | age_group   | total_revenue |
|---|-------------|---------------|
| 1 | Young Adult | 62143         |
| 2 | Midle-aged  | 59197         |
| 3 | Adult       | 55978         |
| 4 | Senior      | 55763         |

## 5. Dashboard in Power BI

The Customer Shopping Behavior Dashboard provides a comprehensive view of customer purchasing patterns, revenue distribution, and key demographic insights. The dashboard is interactive, allowing filtering by subscription status, gender, category, and shipping type.

The dashboard presents a clean, color-coded layout with:

- \*A donut chart showing 27% subscribers vs. 73% non-subscribers\*
- Bar charts displaying revenue distribution across age groups
- Horizontal bar charts comparing revenue by product category
- Sales volume charts showing transaction counts by category
- Filter panels on the left for interactive data exploration



## 6. Business Recommendations

- Increase Subscription Rate (currently 27%) – Offer exclusive discounts or early access to convert non-subscribers.
- Target Young Adults (top revenue segment) – Create age-specific campaigns and product recommendations.
- Strengthen Clothing Category (top performer) – Ensure stock availability and feature Clothing prominently in promotions.
- Address Underperforming Segments – Investigate low Senior revenue with targeted offers or senior-specific products.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.